



Microsoft Azure + Azure DevOps



Exam AZ-104

Microsoft Azure
Administrator Associate



Exam AZ-400

Microsoft Azure DevOps
Engineer Expert

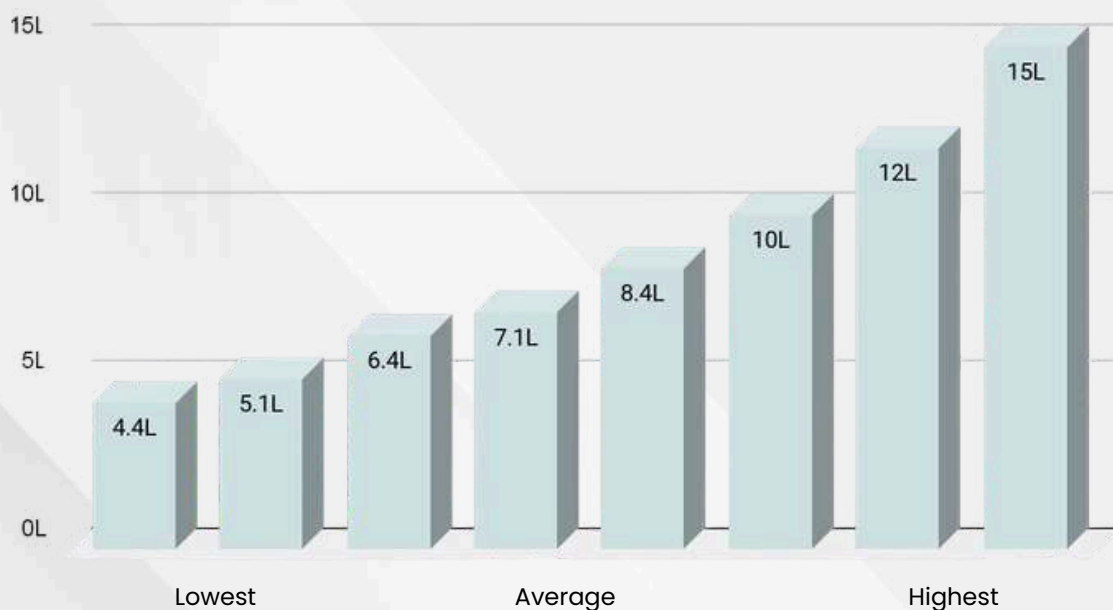
DURATION: 60 Hours

Top Companies Using Azure

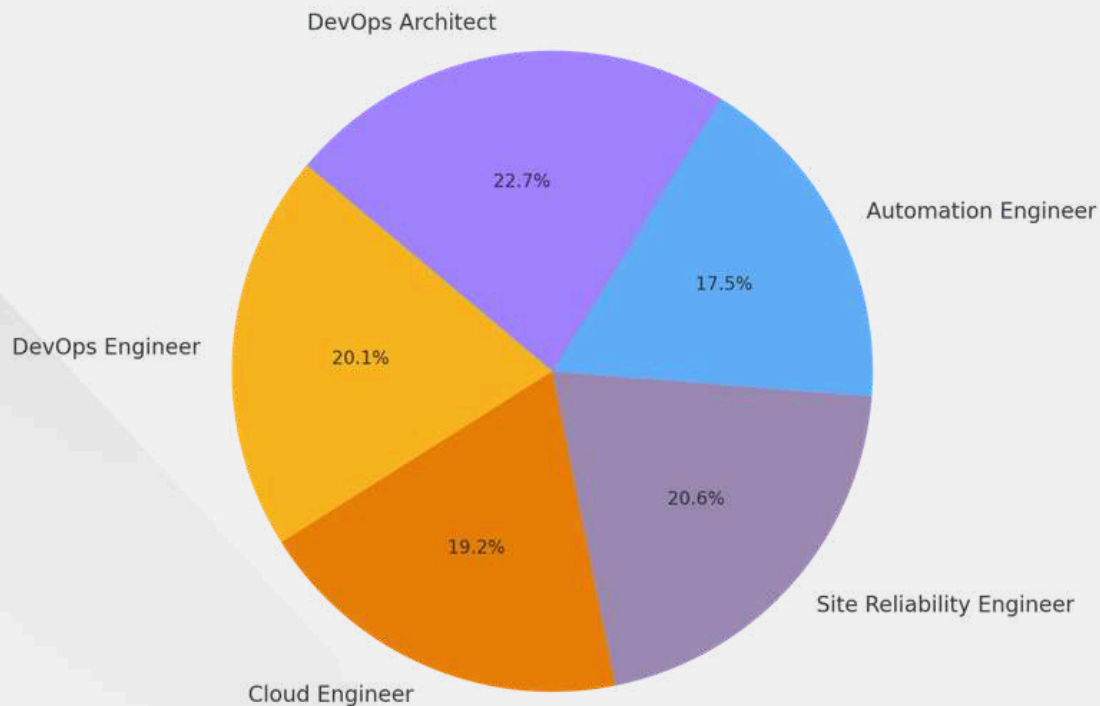
- Azure is a leading cloud platform used by top companies across the globe.
- Major corporations like Adobe, Samsung, and BMW leverage Azure for digital transformation.
- It is a trusted solution for diverse industries, including retail, finance, and healthcare.
- From managing big data to powering AI, Azure helps businesses innovate and scale.



Azure DevOps Engineer Salary Structure



Comparative Salary Growth in DevOps-Related Roles



Why a Career in Azure DevOps is a Smart Move

The global demand for cloud professionals is soaring, and Azure is at the forefront of this growth. A career in Azure DevOps isn't just a job; it's a future-proof investment in your professional growth.

High Demand and Value in the Job Market

Comprehensive and Integrated Platform

Direct Impact on Business Agility and Innovation

Intensive Training to Transform Your Career

Live sessions to gain confidence and to learn by building real-world projects. You will develop an in-depth real-world project by implementing coding workflows.

Learn to manage your project's code with Git and create your project portfolio on GitHub to stand apart from the crowd and demonstrate their skills to potential employers or companies.



Unlock Your Future: Outcomes of Azure Cloud Engineering Program!

- *Transform Your Career Trajectory*
- *Boost Your Earning Potential*
- *Access Elite Opportunities*
- *Become a Leading Azure DevOps Engineer*
- *Master Real-World Application Deployment*
- *Automate with Confidence*
- *Ace Your Technical Interviews*

Approach to Mastering New Skills

Embark on a comprehensive learning adventure designed to transform you from novice to expert. Our platform guides you through four distinct stages, ensuring a deep understanding and practical application of your chosen field.

Learn the Skills



Live Sessions
Video Lessons
Quick Notes

Build Hands-on Exp.



Hands-on Labs
Build Projects

Empower Your Career



Skill Tests
Q&A Courses
Interview Prep.

Trusted by Students for a Reason

Our commitment to your success is reflected in our student reviews. Join the thousands of learners who have achieved their goals with us.



4.8 / 5



**STUDENT
REVIEWS**

4.9 / 5



4.5 / 5



Career & Job Support

Career Counseling

Unlock your true potential with expert guidance.

- Tailored coaching to shape your career path
- Balance personal growth with professional success
- Boost productivity with time-tested strategies
- Build habits that lead to consistent achievement



Project Programs

Stand out with a portfolio that speaks for you.

- Build real projects on GitHub
- Deploy live projects that showcase your expertise
- Create a resume that recruiters notice instantly
- Optimize your profile for maximum job portal visibility



Interview Readiness

Turn interviews into opportunities.

- Prepare for role-specific & technical interviews
- Learn to tackle tough questions with ease
- Build confidence through mock sessions
- Get feedback that makes you interview-ready



Job Guidance

Your success, our mission.

- Dedicated job hunt support
- Personalized guidance every step of the way
- Find roles that truly match your skills



Master Your Exams with Our All-in-One Prep. Program

Practice Tests: **Your Path to Perfection**

You can't pass an exam without practice. We provide a wide range of mock tests that simulate the real exam environment.

- Topic-Based Practice
- Full-Length Mock Exams
- Instant Feedback



Live Sessions: **Learn from the Experts**

Why study alone when you can learn with the best? Our live sessions connect you with experienced teachers who can guide you through every step of your preparation.

- Interactive Classes
- Doubt Clearing
- Expert Strategies



Study Materials: **All the Resources You Need**

We've compiled a complete library of study materials to support your learning journey.

- Comprehensive Notes
- Previous Year Papers
- Important Questions



AZURE CLOUD

Foundation & Prerequisites

- Recap of OS fundamentals and basic Linux commands
- High-level view of networking and server management
- Basic understanding of virtualization concepts

Module: 1 - Cloud Computing Overview

- Why cloud computing emerged (from traditional IT to virtualization to cloud)
- Characteristics and benefits of cloud adoption
- Service models (IaaS, PaaS, SaaS) and deployment models (Public, Private, Hybrid)
- Key cloud providers with a focus on Microsoft Azure
- **Hands-On:** Explore the Azure portal and services catalog.

Module: 2 - Introduction to Azure & Its Architecture

- Overview of Azure and its global infrastructure
- Regions, Availability Zones, and Resource Groups
- Understanding Azure hierarchy (Subscription, Management Groups)
- Introduction to Azure tags and cost management tools
- Using the Azure Pricing Calculator for cost estimates
- **Hands-On:** Create a resource group, explore Azure portal settings, and calculate pricing for a VM.

Module: 3 - Azure Compute Services

- Introduction to Azure Virtual Machines and their architecture
- VM provisioning, sizes, and purchasing options
- Creating and accessing VMs through SSH and RDP
- Custom images and VM storage options
- VM lifecycle management basics
- **Hands-On:** Create a Linux VM and a Windows VM, connect using SSH and RDP, and explore management options.

Module: 4 - Azure Networking Services

- Introduction to Azure Networking and VNets
- Subnets, IP addressing, and basic configurations
- Network Security Groups (NSGs) overview
- Bastion host for secure VM access
- VNet peering overview for multi-VNet connectivity
- **Hands-On:** Create a VNet with subnets, configure NSGs, and connect VMs securely using Bastion.

Module: 5 - High Availability & Scalability

- Concept of high availability and scalability in Azure
- Load Balancing in Azure (Basic vs Standard Load Balancer)
- Auto-scaling concepts and use cases
- Setting up scaling rules
- **Hands-On:** Configure a load balancer for two VMs and test traffic distribution.

Module: 6 - Azure DNS Service

- Role of DNS in Azure
- DNS Zones and DNS records
- Custom DNS configuration in Azure
- **Hands-On:** Create a DNS zone and configure a custom domain for a web app/VM

Module: 7 - Azure CDN

- Role of DNS in Azure
- DNS Zones and DNS records
- Custom DNS configuration in Azure
- **Hands-On:** Create a DNS zone and configure a custom domain for a web app/VM

Module: 8 - Azure Storage Services

- Introduction to Azure Storage and Storage Accounts
- Overview of Blob, File, Queue, and Table storage
- Managing containers and storage lifecycle
- Key features: replication, access tiers, and shared access signatures
- **Hands-On:** Create a storage account, upload blobs, and configure access permissions.

Module: 9 - Azure Database Services

- Concept of high availability and scalability in Azure
- Load Balancing in Azure (Basic vs Standard Load Balancer)
- Auto-scaling concepts and use cases
- Setting up scaling rules
- **Hands-On:** Configure a load balancer for two VMs and test traffic distribution.

Module: 10 - Monitoring & Alerts

- Introduction to Azure Monitor (metrics, logs, insights)
- Configuring alerts and action groups
- Notifications and responses to alerts
- **Hands-On:** Create a monitoring rule and set up an alert with an email notification.

Module: 11 - Azure Entra ID (Identity Services)

- Overview of Azure Entra ID and its benefits
- Core concepts: users, groups, roles
- RBAC (Role-Based Access Control) in Azure
- Introduction to Azure AD Join and AD Connect
- **Hands-On:** Create users and groups, assign RBAC roles, and test access permissions.



Eligible for **Microsoft Azure Administrator Exam (AZ-104)** – opening doors to IT opportunities.

AZURE DEVOPS

✓ Module 1: Introduction to DevOps & CI/CD

- Understanding DevOps workflows & culture.
- CI/CD full lifecycle (Code → Build → Test → Deploy → Monitor).
- Azure DevOps services overview (Repos, Boards, Pipelines, Artifacts).
- **Lab:** Create Azure DevOps Organization, new project setup.

✓ Module 2: Azure Repos – Source Code Management

- Git in Azure Repos (cloning, branching, merging).
- Branch policies, pull requests & approvals.
- Linking repos with external GitHub.
- **Lab:** Create repo, implement branching strategies, enforce PR approvals.

✓ Module 3: Azure Boards – Agile Project Tracking

- Epics, Features, Stories, Tasks.
- Scrum & Kanban workflows.
- Boards integration with Repos & Pipelines.
- **Lab:** Create backlog, sprint plan, and Kanban board.

✓ Module 4: Azure Pipelines – Core CI/CD

- Classic vs YAML pipelines
- Self Hosted & Microsoft Build agents
- Variables, secrets, pipeline environments
- Multi-stage pipelines (CI → CD)
- Pipeline approvals & gated releases

Lab:

- CI pipeline for .NET Core
- Release pipeline to Azure App Service
- Setup pipeline triggers



✓ Module 5: Application Pipelines – Multi-Tech

- .NET Core App: Build, test & deploy to App Service
- Java App: Maven/Gradle build, deploy to Tomcat/Azure Web Apps
- React/Node App: Build & deploy to Azure Static Web Apps/App Service
- API Deployment: REST API pipeline with automated testing
- Artifacts: Manage dependencies using Azure Artifacts

Lab:

- Full CI/CD pipelines for .NET, Java, React apps
- Multi-environment deployments (Dev, QA, Prod)



✓ Module 6: Infrastructure as Code with Pipelines

- Intro to Terraform/ARM templates
- Automating infra deployment in pipelines
- **Lab:** Deploy VNET, Storage, VM/App Service using Terraform/ARM pipeline

✓ Module 7: Containers & Kubernetes with Pipelines

- Build & push Docker images in pipelines
- Deploy container to Azure Web App for Containers
- Intro to Kubernetes (high-level, pre-provided YAML)
- Pipeline for AKS deployment

Lab:

- Dockerize app → Push to ACR → Deploy via pipeline
- Deploy pre-configured YAML to AKS

✓ Module 8: Monitoring, Security & Best Practices

- Pipeline monitoring & logs
- Secrets, approvals, gated deployments
- Pipeline optimization & cost considerations
- **Lab:** Secure pipeline with approvals, policies, and alerts

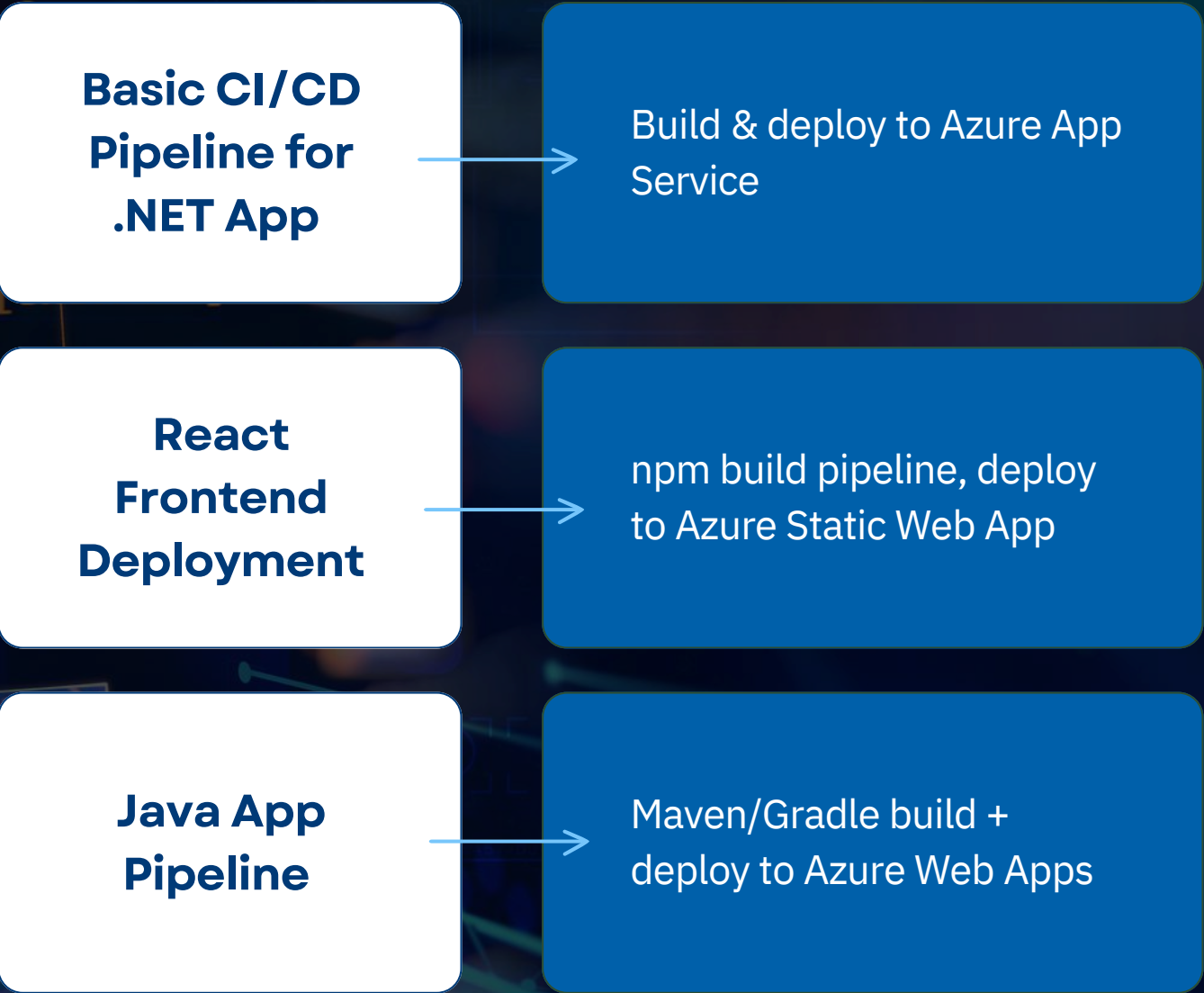


Eligible for **Microsoft DevOps Engineer Expert Exam (AZ-400)** – key to expert-level recognition

Project Portfolio

Students will complete 10 progressive projects (all pipelines they build in class, plus integrated capstones).

Basic CI/CD Pipeline for .NET App



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graph LR; A[Basic CI/CD Pipeline for .NET App] --> B[Build & deploy to Azure App Service]; C[React Frontend Deployment] --> D[npm build pipeline, deploy to Azure Static Web App]; E[Java App Pipeline] --> F[Maven/Gradle build + deploy to Azure Web Apps];
```

Build & deploy to Azure App Service

React Frontend Deployment

npm build pipeline, deploy to Azure Static Web App

Java App Pipeline

Maven/Gradle build + deploy to Azure Web Apps

Multi-Stage Pipeline

CI/CD for Dev → QA → Prod environments

Container Deployment Pipeline

Docker build, push to ACR, deploy to Azure Web App for Containers

IaC + App Deployment Pipeline

Provision infra (ARM/Terraform) + deploy application

Kubernetes Pipeline

CI/CD deploying containerized app to AKS (YAML provided)

Final Capstone Project

End-to-end DevOps workflow integrating Repos, Boards, Pipelines, IaC, Containers, and gated approvals.

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LOCATIONS



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